Fiscal Year:	FY 2004	**Task Last Updated:**	FY 07/05/2004
PI Name:	Lewis, Jennifer A. Sc.D.		
Project Title:	Directed Assembly of Hydroxyapatite Scaffolds		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Urbana	**State:**	IL
Zip Code:	61801	**Congressional District:**	15
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-UNSOL
Start Date:	07/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	Note: Original end date was 12/31/2007.		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Jamison, Russ (University of Illinois)		
Grant/Contract No.:	NAG8-1922		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	PI Lewis will create 3-D periodic hydroxyapatite (HA) scaffolds with controlled porosity on multiple length scales. Concentrated colloidal HA inks have been formulated with controlled viscoelastic properties required for direct-write assembly. 3-D scaffolds have been fabricated via robotic deposition of these inks in a layer-by-layer build sequence. The scaffolds are now being fully characterized with respect to their pore structure at the nanoscale (< 100 nm), microscale (100 nm to 100 microns), and mesoscale (> 100 microns) using both direct imaging techniques and indirect structural probes (e.g., mercury intrusion porosimety). Co PI Jamison will characterize the cellular response of scaffolds of similar design and architecture through in vitro studies. The broad objectives of this ground-based research program are to fabricate and characterize mesoporous hydroxyapatite (HA) structures with microporous domains that serve as model tissue engineering scaffolds. Specifically, we will produce model HA scaffolds via a directed colloidal assembly technique known as robotic deposition. We will then characterize their cellular response through in vitro studies. Our		

	ultimate goal is to develop a fundamental understanding of the interactions between bone tissue and HA scaffolds, and, from that understanding, establish guidelines for the rational design of hydroxyapatite scaffolds that optimize the rate and extent of tissue ingrowth. We expect the knowledge and experience gained in producing diverse architectures of hydroxyapatite to provide valuable insights on producing similar scaffolds from polymers or polymer/HA composites. Such scaffolds could be used not only for bone repair, but for regeneration of skin and other tissues in the body. The versatility, simplicity, and compactness of the robotic deposition equipment used in this research would therefore meet NASA's long-term goal of creating a multifunctional materials platform that can be used as a controllable scaffold for the guided regeneration of a broad spectrum of tissue systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ability to produce engineered, 3-D scaffolds for bone generation could have a profound impact on surgical procedures involving bone grafts, which are currently highly invasive.
Task Progress:	Characterization and Optimization of Hydroxyapatite (HA) Powder (Lewis, Jamison) HA powders were obtained from several commercial sources and characterized to determine their particle morphology, size distribution, specific surface area, crystallinity, and chemical composition. In all cases, the as-received powders were comprised of microporous, high specific surface area (> 70 m2/g) agglomerates. To create dense powders suitable for producing concentrated HA inks for direct writing of 3-D periodic scaffolds, these powders had to be calcined and milled. Through careful optimization of calcining temperature and ball milling conditions, a dense powder with average diameter of 1.5 μm and specific surface area of 6.9 m2/g were produced. Development of Concentrated HA Inks (Lewis, Jamison) The successful fabrication of 3-D periodic scaffolds using robotic deposition depends critically upon the development of concentrated HA inks (~ 48% by volume) with controlled viscoelastic properties. First, stable HA suspensions were produced by dispersing calcined/milled HA powder in an aqueous suspension using a polyelectrolyte dispersant. Next, the ink was gelled by either changing pH or ionic strength to achieve the desired rheological response. The apparent viscosity and elastic modulus of these inks were characterized as a function of shear stress, and the gel strength was tuned to facilitate both ink flow through fine deposition nozzles (D = 250 – 750 microns) as well as to provide shape retention of the as-extruded filaments during the assembly process. Selection of porogens (Jamison) Polymethylmethacrylate (PMMA) microspheres of average diameter 6.1 μm were obtained from a commercial source. These microspheres serve as porogens yielding microscopic pores within the scaffold upon burnout at elevated temperature. The optimal PMMA loading was determined to be 27.5 vol% HA and 27.5 vol% PMMA microspheres, which resulted in the formation of interconnected micropores of average diameter in the final sintered structures. 3-D Scaffold Fabrication (Lewis, Jamison) 3-D periodic scaffolds with varying geometry and HA volume fraction were created by direct-write assembly using a 3-axis, motion-controlled robotic deposition stage. The ink reservoir is mounted on the z-axis, and the ink is dispensed through a fine cylindrical nozzle at an appropriate volumetric flow rate required to allow patterning at writing speeds of ~ 5 mm/s. During assembly, an ink filament is extruded continuously and patterned in a layer-by-layer build sequence until the entire 3-D structure is produced. HA scaffolds with radial, concentric, and rectilinear geometries have been fabricated including designs with gradient dimensions. The spacing between filaments (or rods) varied between 150 to 700 μm within these scaffolds. In addition to uniform structures, a series of quadrant parts were also created that allowed multiple rod configurations to be probed simultaneous in a single scaffold structure. 3-D HA scaffolds were created with multiscale porosity by varying rod spacing and geometry, porogen content, and sintering conditions. The largest pore channels (~ 150 – 750 microns), defined by the rod geometry and lattice constant, are interconnected in all three dimensions within the scaffold. The intermediate pore channels (~ 5 microns), formed by porogen removal, are also interconnected in all three dimensions within the individual filamentary elements of the scaffold structure. Finally, the finest pore channels (< 1 micron), developed by incomplete sintering, can either be in the form of interconnected or closed porosity depending on the sintered density. Mechanical properties of materials and scaffolds (Jamison) The use of HA in bone replacement is currently limited to non-load bearing applications, because of the relatively low strength of available forms of the material compared to natural bone. One of the opportunities presented by the high controllability inherent in direct-write assembly is an increase in strength of the resultant HA structure. Cylindrical specimens for compression testing and model HA scaffolds were fabricated and tested to establish the upper and lower limits of strength and stiffness. The bulk HA exhibits an average compressive strength of 766 MPa and average compressive modulus of 30 GPa. These properties compare favorably to properties of human cortical bone which has typical compressive strength 195 MPa and compressive modulus of 17 GPa. In contrast, the average compressive strength of 3-D periodic HA scaffolds was 137 MPa. This value reflects the presence of large-scale interconnected porosity coupled with the dominant mode of (tensile) failure of the HA struts within these scaffolds. Bulk HA with included microporosity (porogens) exhibited an average compressive strength and modulus values of 186 MPa and 19 GPa, respectively. In contrast, 3-D periodic HA scaffolds with included microporosity exhibited an average compressive strength of 43 MPa. This represents a nearly three-fold reduction from dense HA scaffolds. Scaffolds possessing these mechanical properties would be suitable for use in some types of facial reconstruction and other minimal load bearing applications. Imaging cells on scaffolds (Jamison) Rapid and complete cell and tissue ingrowth into a tissue engineered scaffold is a critical factor in achieving full bone remodeling. It is necessary therefore to assess the temporal and spatial distribution of cells within the pores of the structure when comparing specific design options. Conventional methods of section histology can be slow and labor-intensive. For HA scaffolds section histology is particularly challenging because of the inherent hardness and brittleness of the material. Consequently an alternative, non-destructive method of performing this assessment was investigated. Scaffolds were seeded with mouse mesenchymal stem cells and cultured for periods of days or weeks in vitro. These cell-scaffold constructs were removed from culture, rinsed, fixed with glutaraldehyde, and post-fixed with osmium tetroxide (OT). When examined using micro-computed tomography (μ-CT) the cells were distinguishable from the scaffold to which they were attached by the radiopacity of the OT they had absorbed during post-fixation. Although the resolution of commercially available μ-CT instruments is on the order of individual cell dimensions (approximately 10 μm), it was adequate to assess the spatial location of cell clusters within samples of scaffold having overall dimensions of several millimeters. Using higher resolution instruments at Argonne National Laboratory and working with collaborators at University of Illinois,

	Argonne, and Northwestern University, individual cells on HA scaffolds have been imaged. The refinement of staining methods and the use of this technique to distinguish extracellular matrix components are the subject of continuing research. Custom mandible implant (Jamison) One of the long-term objectives of this project is to establish design rules for the fabrication of load-bearing scaffolds to restore bone lost to trauma or disease. It is conceivable that such implants could be designed for a specific site in an individual patient using existing computer tools for automated design and manufacturing. While the tasks in this first year of the project have largely focused on processing and fabrication issues at the microscopic level, we undertook as a proof of concept task, the development of a hydroxyapatite scaffold designed for restoration of a mandibular defect in a human patient. Beginning with the computed tomography (CT) scan of the patient, a computer model of the patient's mandible was created. Working with the patient's attending physician, a maxillofacial surgeon, the design of a graphical model of the implant required to fill the bone defect was created. The graphical model was converted to a computer aided manufacturing model and sent electronically to collaborators at Sandia National Laboratories. There a block HA scaffold was fabricated using the directed deposition method. The block was infused with wax and a replica of the bone defect was machined from the block. The wax was removed from the implant by heating to leave a hydroxyapatite implant of a size and shape of the surgeon's original design. This represents only a proof of concept outcome because such implants are not yet approved for human implantation by the U.S. Food and Drug Administration. It does demonstrate however the feasibility of designing and fabricating custom hydroxyapatite bone scaffolds using existing modeling tools and fabrication approaches developed in this project. A description of the work is accessible at www.itg.uiuc.edu/technology/reconstruction/.
Bibliography Type:	Description: (Last Updated: 12/21/2007)
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